

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Civil) (Credit System) Degree Examinations**

May - June 2023

20CV603 - TRANSPORTATION ENGINEERING - II

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Assume any missing data suitably as per Indian Standards.

Unit – I**Marks BT* CO* PO***

- | | | | | | |
|-------|--|----|-----|---|---|
| 1. a) | (i) Explain the role of railways in the development of any country.
(ii) Enumerate any two functions and four requirements of ideal sleepers. | 10 | L*2 | 1 | 1 |
| b) | Define negative super-elevation. A branch line of 5° branches out from a 3° main curve in an opposite direction in a layout of a MG yard. If the speed of the branch line is 40 kmph. Calculate the permissible speed on the main line allowing the maximum cant deficiency. | 10 | L3 | 2 | 2 |
| 2. a) | Determine the quantities of materials as per IRB guidelines which are required to construct a 3 kms of railway M.G. permanent way using 90 R rails. Assume the sleeper density as N+3. | 10 | L3 | 1 | 1 |
| b) | Write a note on (i) Ruling gradient (ii) Grade compensation. Indicate IRB specifications. | 10 | L2 | 2 | 1 |
| 3. a) | Explain the concept of creep in railways. List any four factors on which the extent of creep depends upon. Explain wave action theory of creep with a neat sketch. | 10 | L2 | 1 | 1 |
| b) | Compute the steepest gradient that a train of 25 goods wagons and a locomotive can haul given the following data:
Weight of each goods compartment on axle = 22 t
Weight of Railway Engine = 190 t
Tractive effort of engine = 18 t
Rolling resistance of engine = 4 kg/t
Rolling resistance of following wagons = 3.5 kg/t
Assume the average speed of this goods train as 70 kmph. | 10 | L3 | 2 | 2 |

Unit – II

- | | | | | | |
|-------|---|----|----|---|---|
| 4. a) | Briefly explain the steps involved in following Tunneling methods with the relevant sketches (i) Needle-Beam Method (ii) Drift method. | 10 | L2 | 3 | 1 |
| b) | List and explain any five aircraft characteristics which govern the airport design. | 10 | L2 | 4 | 2 |
| 5. a) | With a neat sketch, explain (i) semaphore signal. (ii) Marshalling yard. | 10 | L2 | 3 | 1 |
| b) | Sketch typical plan and cross-section of an ILS runway and explain any three geometric design elements of runway. Indicate the ICAO specifications. | 10 | L2 | 4 | 2 |
| 6. a) | What do you mean by a Railway station? With the help of sketches explain Class A and Class B railway Station? | 10 | L2 | 3 | 1 |

P.T.O.

- b) The length of runway under standard condition is 1620 m. The airport site has an elevation of 270m. Its reference temperature is 36.94°C. If the runway is to be constructed with an effective gradient of 0.2%, determine the corrected runway length.

10 L3 4 1

Unit – III

7. a) Explain the following components of artificial harbors (i) Quays (ii) Wharves (iii) Piers (iv) Dolphins.
- b) Design an exit taxiway joining a runway and a parallel main taxiway. The total angle of turn is 35° and the maximum turn-off speed is 95 kmph. Suggest the maximum separation clearance and draw a neat sketch showing all the design elements. Assume all the missing data suitably as per ICAO/FAA guidelines and indicate clearly the same at the adequate place.
8. a) Explain any five geometric design elements of a taxiway. Indicate ICAO recommendations.
- b) With sketches explain (i) Natural Classification of harbor (ii) Types of breakwaters (any two).

10 L2 5 1

10 L3 5 2

10 L2 5 1

10 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
